Supplementary Material for "Short-Observation Seven-Feature Detection of Small Floating Targets in Sea Clutter Under Record-Disjoint Evaluation"

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This document collects the ablation figure of the first design, its sensitivity and audit tables (S1–S7), and the duration, equal-dwell, calibration, and shift tables of the final detector (S8–S11), the first-design results moved from the manuscript (Figs. S2–S3 and Table S12), and the IPIX1998 record-by-seed results of the final detector (Table S13). All values are recomputed from the per-run files in the supplementary archive; table and figure numbers carry the prefix S.

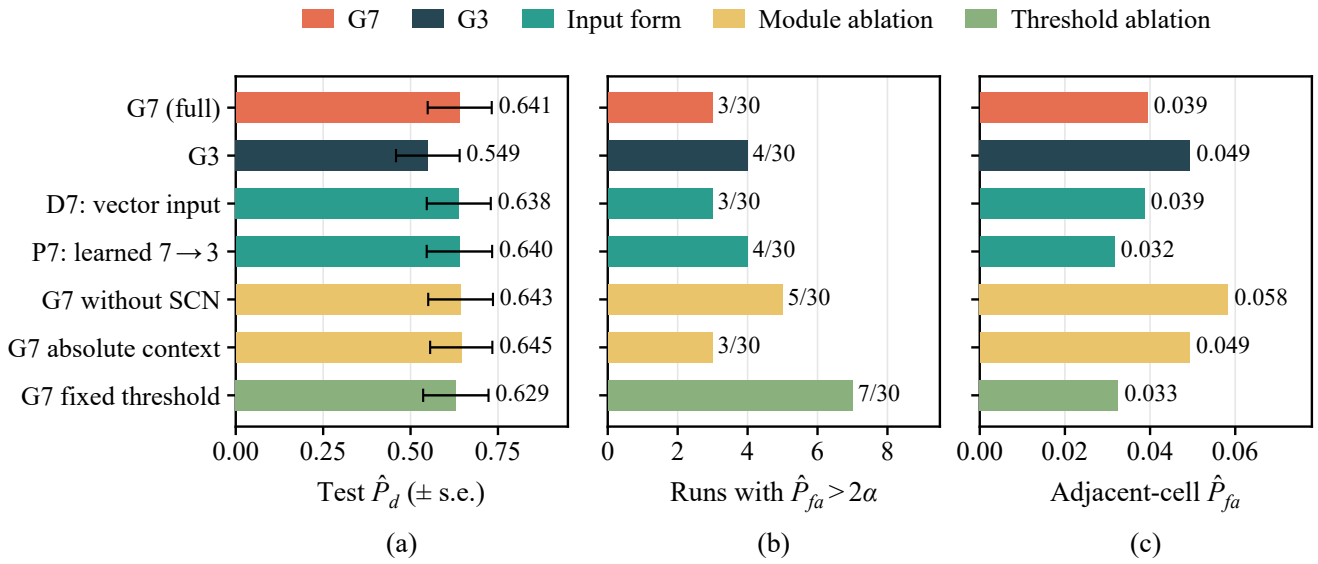


Fig. S1. Arms at $\alpha = 10^{-2}$ (IPIX1993 LORO, tail CFAR unless stated). (a) Test P_d, mean and standard error over the ten fold means. (b) Runs, of 30, with test $P_{fa} > 2\alpha$. (c) Adjacent-cell P_{fa}.

TABLE S1
PARAMETERS AND PER-CELL HEAD INPUT WIDTH OF EVERY NETWORK ARM.

Arm	Input	Head input	Parameters
G7	SA-RGB, 7 descriptors	368	27 841
G3	SA-RGB, 3 descriptors	368	27 841
D7	7-vector per cell	218	18 241
P7	learned 7 → 3	98	10 582
G7 w/o SCN	SA-RGB, 7 descriptors	368	27 841
G7 abs. context	SA-RGB, 7 descriptors	332	25 537
G7 range-ordered	SA-RGB, 7 descriptors	72	8 897

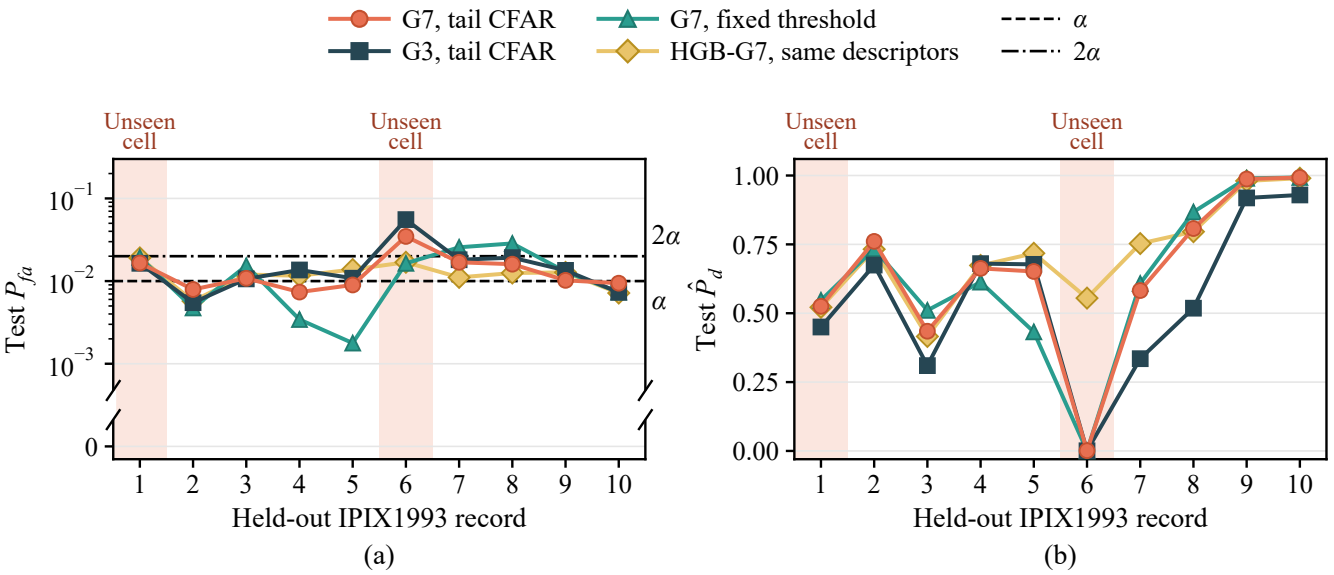


Fig. S2. First design, per-fold IPIX1993 LORO results at $\alpha = 10^{-2}$ (mean over three seeds; the convex hull is deterministic). (a) Realized test P_{fa}; $P_{fa} = 0$ is drawn below the axis break. (b) Test P_d. Shaded: folds 1 and 6, whose target cell index is absent from the training targets.

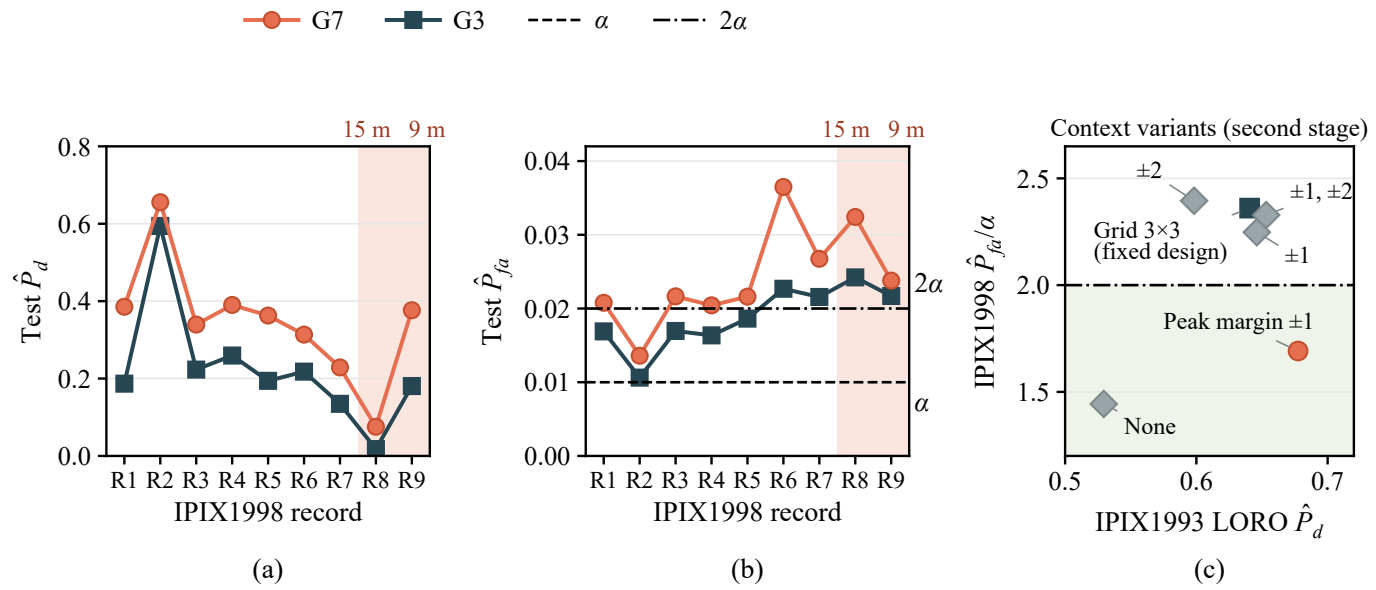


Fig. S3. First design on IPIX1998 at $\alpha = 10^{-2}$, trained and calibrated on IPIX1993. (a) Test P_d per record; shaded: R8 (15 m) and R9 (9 m). (b) Realized test P_{fa} per record. (c) Second stage: IPIX1993 LORO P_d against IPIX1998 P_{fa}/α for G7 with different context inputs to the cell head (single observations); green: within 2α.

TABLE S2
TAIL-CFAR SETTINGS (IPIX1993 LORO, GRID-DIRECTION NETWORK). W: REFERENCE FRAMES; TRIM: HIGHEST CELLS OMITTED PER REFERENCE FRAME.

Arm	W	Trim	$\hat{P}_d$	$\hat{P}_{fa}/\alpha$	Runs $> 2\alpha$	$\hat{P}_d$	$\hat{P}_{fa}/\alpha$	Runs $> 2\alpha$
G7	64	1	0.641	1.39	3/30	0.455	2.94	9/30
G7	32	1	0.637	1.38	3/30	0.453	3.19	9/30
G7	16	1	0.632	1.38	4/30	0.452	3.53	12/30
G7	64	2	0.622	1.71	5/30	0.462	7.40	12/30
G7	32	2	0.615	1.66	5/30	0.459	7.44	12/30
G3	64	1	0.549	1.71	4/30	0.419	5.65	20/30

Columns 4–6: $\alpha = 10^{-2}$; columns 7–9: $\alpha = 10^{-3}$. The design uses $W = 64$, trim 1.

TABLE S3
REALIZED-TO-NOMINAL RATIO $\hat{P}_{fa}/\alpha$ OF THE TAIL CFAR WITH c SET AT $\alpha = 10^{-2}$ (IPIX1993 LORO, MEAN OVER 30 RUNS).

Arm	$\alpha = 0.1$	$\alpha = 0.03$	$\alpha = 0.01$	$\alpha = 0.003$	$\alpha = 0.001$
G7	0.88	1.23	1.39	1.78	2.94
G3	0.74	1.13	1.71	2.98	5.65

A ratio that grows as α falls indicates a score tail heavier than the exponential model.

TABLE S4
TAIL AND FIXED RULES ON THE COMMON 1020 FRAMES PER RECORD ($\alpha = 10^{-2}$, IPIX1993 LORO).

Arm	Rule	$\hat{P}_d$	$\hat{P}_{fa}/\alpha$	Adjacent $\hat{P}_{fa}$	Background $\hat{P}_{fa}/\alpha$
G3	fixed	0.539	1.41	0.033	1.06
G3	tail	0.549	1.71	0.049	1.12
G7	fixed	0.630	1.36	0.033	1.01
G7	tail	0.641	1.39	0.039	0.92

Warm-up frames (fewer than 64 frames of history) versus steady state, tail rule: G3 1.60α versus 1.74α; G7 1.31α versus 1.41α.

TABLE S5
NESTED SELECTION WITHIN THE HGB DEVELOPMENT PIPELINE. D1: NORMALIZATION; D2: CONTEXT; D3: THRESHOLD RULE (TEXP: EXPONENTIAL TAIL; CMAP: EMPIRICAL CLUTTER MAP).

Outer fold	D1	D2	D3	$\hat{P}_d$ (10^{-2})	$\hat{P}_{fa}/\alpha$	$\hat{P}_d$ (10^{-3})	$\hat{P}_{fa}/\alpha$
1	frame	rel	texp	0.522	1.91	0.354	3.5
2	frame	rel	cmap	0.721	0.54	0.716	5.3
3	frame	rel	cmap	0.421	1.27	0.397	11.3
4	frame	rel	texp	0.674	1.16	0.521	1.9
5	frame	rel	cmap	0.682	1.12	0.622	6.9
6	frame	rel	cmap	0.535	1.79	0.441	15.5
7	frame	rel	cmap	0.726	1.03	0.667	6.4
8	frame	rel	texp	0.796	1.25	0.621	1.7
9	frame	rel	cmap	0.980	1.34	0.965	3.2
10	frame	rel	cmap	0.989	0.73	0.975	1.9

TABLE S6
CONVEX-HULL REFERENCE: PEELING LAYER, CALIBRATION FEASIBILITY, AND TEST RESULTS.

Protocol	α	Fold	Layer	Feasible	Val. P_{fa}	$\hat{P}_d$	$\hat{P}_{fa}$
LORO	0.01	1	0	yes	0.0098	0.104	0.0254
LORO	0.01	2	2	yes	0.0080	0.152	0.0256
LORO	0.01	3	1	yes	0.0059	0.021	0.0033
LORO	0.01	4	0	yes	0.0063	0.062	0.0038
LORO	0.01	5	0	yes	0.0040	0.014	0.0028
LORO	0.01	6	2	yes	0.0065	0.245	0.0247
LORO	0.01	7	14	yes	0.0081	0.276	0.0718
LORO	0.01	8	5	yes	0.0085	0.034	0.0005
LORO	0.01	9	2	yes	0.0092	0.220	0.0020
LORO	0.01	10	4	yes	0.0061	0.276	0.0177
IPIX1998	0.01	–	2	yes	0.0089	0.505	0.0394
LORO	0.001	1	0	no	0.0101	0.106	0.0201
LORO	0.001	2	0	no	0.0014	0.080	0.0083
LORO	0.001	3	0	no	0.0035	0.013	0.0014
LORO	0.001	4	0	no	0.0041	0.035	0.0031
LORO	0.001	5	0	no	0.0077	0.098	0.0059
LORO	0.001	6	0	no	0.0016	0.145	0.0070
LORO	0.001	7	4	yes	0.0007	0.097	0.0131
LORO	0.001	8	1	yes	0.0003	0.003	0.0002
LORO	0.001	9	0	no	0.0013	0.076	0.0002
LORO	0.001	10	1	yes	0.0002	0.128	0.0030
IPIX1998	0.001	–	0	no	0.0024	0.384	0.0226

Feasible: a peeling depth reaches a validation rate of at most α; otherwise the unpeeled outer hull (layer 0) is used.

TABLE S7
MATCHED-INPUT COMPARISON: IDENTICAL 72-CHANNEL RANGE-ORDERED TENSOR GIVEN TO RADARGRID-CS AND HGB.

Scorer	Protocol	Rule	α	$\hat{P}_d$	$\hat{P}_{fa}/\alpha$	Runs $> 2\alpha$
HGB	IPIX1998	fixed	0.001	0.351	8.66	1/1
HGB	IPIX1998	fixed	0.01	0.534	2.78	1/1
HGB	IPIX1998	oracle	0.001	0.118	–	–
HGB	IPIX1998	oracle	0.01	0.372	–	–
HGB	IPIX1998	texp	0.001	0.198	4.39	1/1
HGB	IPIX1998	texp	0.01	0.425	1.53	0/1
HGB	LORO	fixed	0.001	0.570	2.66	3/10
HGB	LORO	fixed	0.01	0.724	1.61	2/10
HGB	LORO	oracle	0.001	0.529	–	–
HGB	LORO	oracle	0.01	0.700	–	–
HGB	LORO	texp	0.001	0.509	1.52	2/10
HGB	LORO	texp	0.01	0.701	1.18	0/10
network	IPIX1998	fixed	0.001	0.328	8.43	3/3
network	IPIX1998	fixed	0.01	0.555	3.13	3/3
network	IPIX1998	oracle	0.001	0.112	–	–
network	IPIX1998	oracle	0.01	0.355	–	–
network	IPIX1998	texp	0.001	0.268	5.87	3/3
network	IPIX1998	texp	0.01	0.429	1.69	0/3
network	LORO	fixed	0.001	0.533	2.53	12/30
network	LORO	fixed	0.01	0.727	1.72	12/30
network	LORO	oracle	0.001	0.485	–	–
network	LORO	oracle	0.01	0.689	–	–
network	LORO	texp	0.001	0.514	2.01	14/30
network	LORO	texp	0.01	0.678	1.13	0/30

oracle: P_d at equal realized test $P_{fa} = \alpha$ (separability, not deployable). texp: tail CFAR; fixed: validation threshold. IPIX1998 G7 (grid direction), pooled versus record-averaged: $\hat{P}_d$ 0.374 versus 0.348, $\hat{P}_{fa}/\alpha$ 2.36 versus 2.42; G3: 0.247 versus 0.223, 1.84 versus 1.88.

TABLE S8
FINAL DETECTOR ACROSS OBSERVATION DURATIONS (IPIX1993 LORO, G7, TAIL CFAR, $\alpha = 10^{-2}$): $\hat{P}_d$ ($\hat{P}_{fa}/\alpha$; RUNS $> 2\alpha$).

d (s)	$K = 1$	$K = 2$	$K = 4$	$K = 8$
0.256	0.654 (1.13; 0/30)	0.727 (1.12; 0/30)	0.798 (1.16; 2/30)	0.852 (1.09; 1/30)
0.512	0.685 (1.13; 0/30)	0.763 (1.11; 0/30)	0.829 (1.08; 0/30)	0.872 (1.16; 0/30)
1.024	0.702 (1.08; 0/30)	0.771 (1.05; 0/30)	0.839 (1.10; 0/30)	0.885 (1.22; 4/30)

TABLE S9
EQUAL DWELL WITH MATCHED TRAINING FRAMES (121 PER SEQUENCE; IPIX1993 LORO, G7, TAIL CFAR, $\alpha = 10^{-2}$).

Dwell (s)	d (s)	K	Train frames	$\hat{P}_d$	$\hat{P}_{fa}/\alpha$
1.024	0.256	4	121	0.780	1.12
1.024	0.512	2	121	0.748	1.09
1.024	1.024	1	121	0.702	1.08
2.048	0.256	8	121	0.834	1.04
2.048	0.512	4	121	0.815	1.11
2.048	1.024	2	121	0.771	1.05

TABLE S10
TAIL-CFAR CALIBRATION OF THE FINAL DETECTOR: $\hat{P}_{fa}/\alpha$ WITH c SET AT 10^{-2} (G7, $d = 0.512$ S).

Protocol	K	$\alpha = 0.1$	$\alpha = 0.03$	$\alpha = 0.01$	$\alpha = 0.003$	$\alpha = 0.001$
IPIX1998	1	0.67	1.04	1.68	2.87	4.92
IPIX1998	4	0.69	0.91	1.17	1.68	2.51
IPIX1993 LORO	1	0.74	0.95	1.13	1.42	1.88
IPIX1993 LORO	4	0.90	1.11	1.08	0.88	0.78

TABLE S11
CYCLIC RANGE-SHIFT TEST OF THE FINAL DETECTOR (IPIX1993 LORO, G7, $K = 4$, TAIL CFAR, $\alpha = 10^{-2}$).

Shift s	$\hat{P}_d$	$\hat{P}_{fa}/\alpha$
-2	0.825	1.08
-1	0.827	1.09
0	0.829	1.08
1	0.828	1.13
2	0.825	1.10

TABLE S12
FIRST DESIGN AND REFERENCE DETECTORS, IPIX1993 LORO AT $\alpha = 10^{-3}$ (DESCRIPTIVE; TAIL CFAR CALIBRATED AT 10^{-2} UNLESS STATED).

Arm	$\hat{P}_d$	$\hat{P}_{fa}/\alpha$	Runs $> 2\alpha$
G7	0.455	2.9	9/30
G3	0.419	5.6	20/30
D7	0.460	2.9	8/30
P7	0.453	2.1	11/30
G7 w/o SCN	0.451	3.2	6/30
G7 abs. context	0.462	4.0	8/30
G7 fixed threshold[a]	0.465	1.7	10/30
Amplitude ratio (RAA)	0.028	0.6	1/10
Convex hull[a,b]	0.078	6.2	7/10
HGB-G3	0.434	3.2	5/10
HGB-G7	0.524	1.5	3/10
Range-ordered RadarGrid-CS	0.514	2.0	14/30
Range-ordered HGB	0.509	1.5	2/10

[a]Threshold calibrated on the validation record at 10^{-3}. [b]No peeling depth reaches a validation rate of at most 10^{-3} in 7 of 10 folds; the unpeeled outer hull (layer 0) is then used, so these entries are not calibrated operating points.

TABLE S13
FINAL DETECTOR ON IPIX1998 PER RECORD AND SEED: $\hat{P}_d$ / $\hat{P}_{fa}/\alpha$ (G7, $K = 4$, TAIL CFAR, $\alpha = 10^{-2}$).

Record	Seed 0	Seed 1	Seed 2
R1	0.750 / 0.65	0.737 / 0.93	0.808 / 0.98
R2	0.918 / 0.76	0.922 / 0.74	0.924 / 0.57
R3	0.682 / 0.95	0.678 / 0.95	0.675 / 0.74
R4	0.614 / 0.84	0.599 / 0.90	0.576 / 1.01
R5	0.474 / 0.61	0.456 / 0.81	0.431 / 0.68
R6	0.253 / 1.87	0.196 / 1.80	0.259 / 2.06
R7	0.607 / 1.12	0.614 / 0.78	0.644 / 0.97
R8	0.313 / 2.04	0.277 / 1.79	0.299 / 2.11
R9	0.472 / 2.21	0.490 / 2.01	0.560 / 1.60

Decision frames 8–116 of each window. Five of the 27 record–seed rates exceed 2α, all on R6, R8, and R9; pooled over the nine records the rate is 1.15–1.18α per seed.